

Blood Harmane (1-methyl-9h-pyrido[3,4-b]indole) Concentrations in Essential Tremor: Repeat Observation in Cases and Controls in New York

Essential tremor (ET) is a widespread late-life neurological disease. Genetic and environmental factors are likely to play important etiological roles. Harmane (1-methyl-9H-pyrido[3,4-b]indole) is a potent tremor-producing neurotoxin. Previously, elevated blood harmane concentrations were demonstrated in ET cases compared to controls, but these observations were all been cross-sectional, assessing each subject at only one time point. Thus, no one has ever repeat-assayed blood harmane in the same subjects twice. Whether the observed case-control difference persists at a second time point, years later, is unknown. The current goal was to re-assess a sample of our ET cases and controls to determine whether blood harmane concentration remained elevated in ET at a second time point. Blood harmane concentrations were quantified by a well-established high performance liquid chromatography method in 63 ET cases and 70 controls. A mean of approximately 6 years elapsed between the initial and this subsequent blood harmane determination. The mean log blood harmane concentration was significantly higher in cases than controls (0.30 ± 0.61 g–10/ml vs. 0.08 ± 0.55 g–10/ml), and the median value in cases was double that of controls: 0.22 g–10/ml vs. 0.11 g–10/ml. The log blood harmane concentration was highest in cases with a family history of ET. Blood harmane concentration was elevated in ET cases compared to controls when re-assessed at a second time point several years later, indicating what seems to be a stable association between this environmental toxin and ET.